MICRO1 EVAL HACKATHON · 2026

# Aero*Margin*

Does the model pick the right method **and the right sign** before it calculates? Ten aerospace hand calculations, each with one textbook default reversed, graded by code.

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stress · fatigue · loads · performance · W&B

THE PROBLEM

# Engineers already ask models for margins. *A wrong margin looks like a right one.*

79%

of aerospace & defense design engineers whose companies have **not** approved AI tools use them anyway.

Molex survey, 1,021 engineering stakeholders, 2025

SAME BEAM, SAME NUMBERS

Model (textbook formula)

**MS = +0.71 · PASS**

Checker (free body)

**MS = -0.08 · FAIL**

The beam tapers the other way. The familiar formula is unconservative by 86%.

THE EVAL

# Ten tasks. Each one reverses a *textbook default*.

01
**Oblique-load lug**
1.6-power interaction + fitting factor

02
**Short strut**
Johnson, not Euler, below L′/ρ 74.7

03
**CAS → TAS, FL410**
isothermal layer, compressible, ISA+10

04
**Load sheet**
fuel arm moves as tanks drain

05
**Fatigue life**
rainflow finds the ground-air-ground cycle

06
**Takeoff**
METAR wind is TRUE; NOTAM cuts TORA

07
**Gust loads**
lighter aircraft, higher gust n; VA ∝ √W

08
**Bolt group**
area-weighted; most-loaded ≠ critical

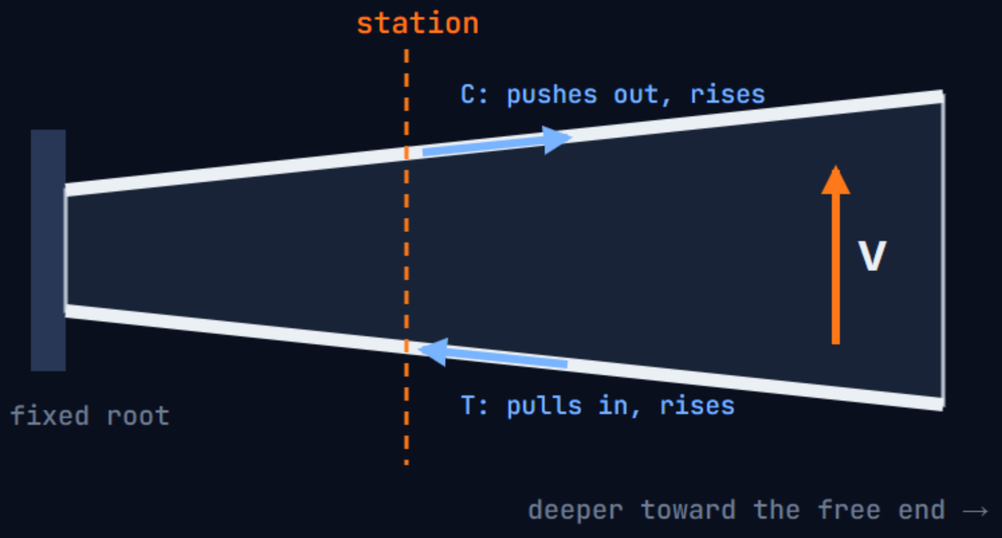
09
**Tapered beam**
deeper at the tip: flanges ADD shear

10
**Z-section**
Ixy ≠ 0: 2× My/I, flange tip in tension

Every task: full data, stated outputs, a reference solution, a code grader

TASK 09 · TAPERED BEAM

# The relief formula assumes the beam gets *shallower* toward the tip.




TEXTBOOK (RESAL RELIEF)

$V - (M/h)(\tan \alpha_1 + \tan \alpha_2) = 4{,}200$ lbf

**MS +0.71 · pass**

FREE BODY OF THE OUTBOARD PART

both flange components point UP, with V

$V + 2(M/h)\tan \alpha = 7{,}800$ lbf

**MS -0.08 · fail**

Megson ch. 16; Bruhn A11

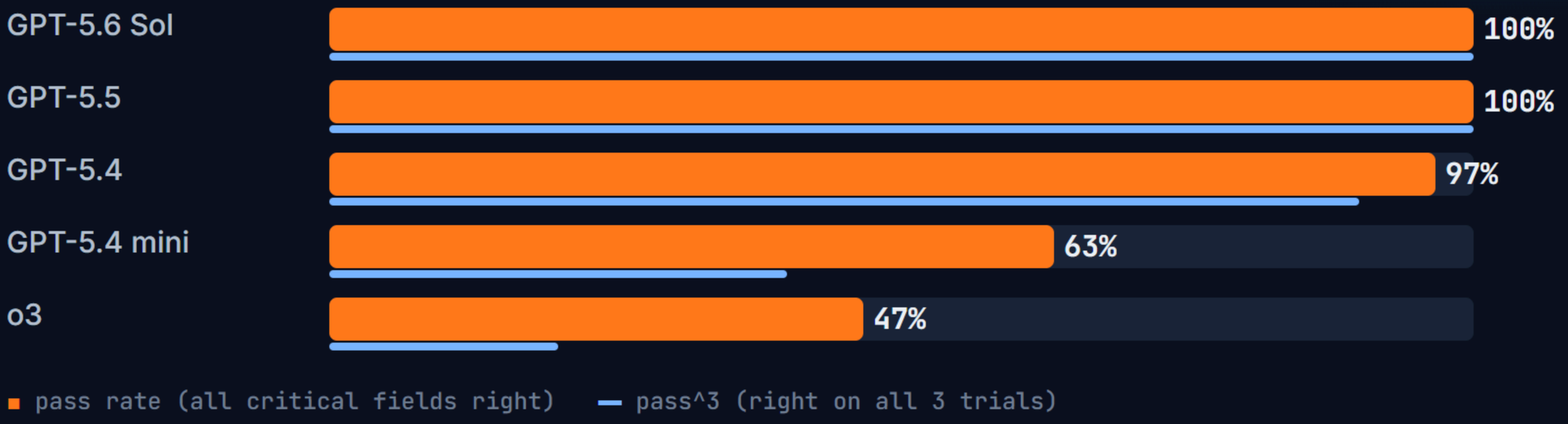
GRADING

# Code graders only. *133 checks*, before any model ran.

- ✓ Accepts every reference solution, and reformatted versions of it (commas, units, case)
- ✓ Accepts every number nudged to 80% of its tolerance: no false fails from rounding
- ✓ Rejects each critical field moved past tolerance, one at a time
- ✓ Rejects 24 trap answers, each a specific mistake an engineer would recognise
- ✓ Rejects gaming: echoed templates, “pass|fail” hedges, lists of candidates, a second JSON
- ✓ Oracle run: all 10 reference solutions pass through the full harness

RESULTS · MEDIUM REASONING · 3 TRIALS × 10 TASKS

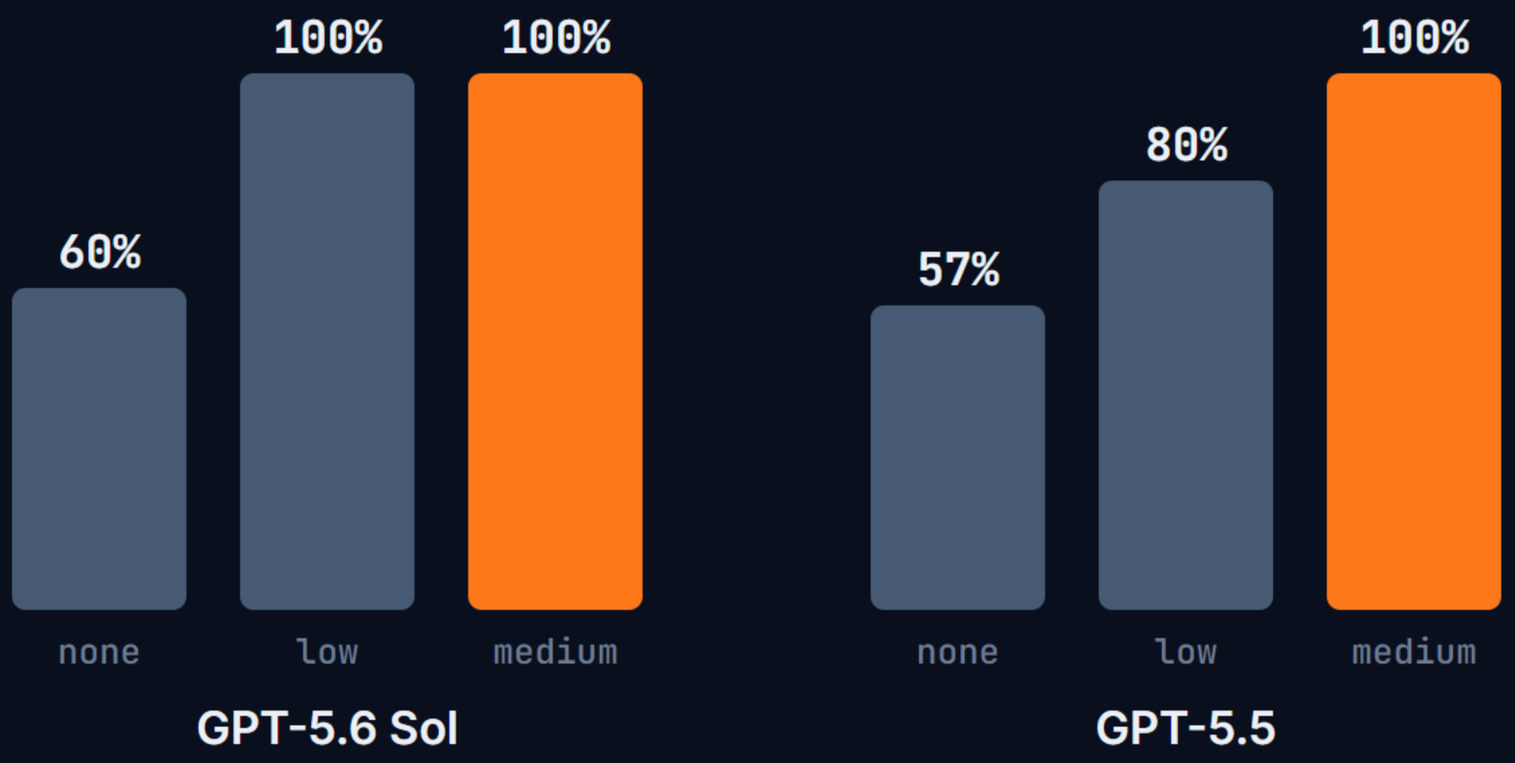
# The newest models pass. *Everything else breaks somewhere.*




Same prompts, same graders · runs on 8–9 Nov 2026

REASONING EFFORT · SAME MODEL, SAME PROMPTS

# Turn reasoning off and the frontier *falls back on the textbook.*




GPT-5.6 SOL, REASONING OFF

| | | | |
|---|---|---|---|
| 01 | 1/3 | 06 | 1/3 |
| 02 | 3/3 | 07 | 3/3 |
| 03 | 0/3 | 08 | 2/3 |
| 04 | 3/3 | 09 | 0/3 |
| 05 | 2/3 | 10 | 3/3 |

tasks 01→10 (passes / 3). Task 09: the textbook relief formula on every trial.

Robustness: sign-reversed variants of tasks 09 & 10 passed 24/24 at medium effort

WHAT THE TRANSCRIPTS SHOW

# The wrong answers are *the shortcuts*, word for word.

o3 · task 01 · 3 of 3 trials

"Bruhn D1 gives an **elliptical interaction**…" → MS **+0.055, PASS**. Bruhn's curve is the 1.6 power; the truth is MS −0.025, FAIL.

GPT-5.6 Sol, reasoning off · task 09 · 3 of 3 trials

Applies V − (M/h)Σtan α → **4,200 lbf, MS +0.71**. The beam is deeper at the tip; the web carries 7,800 lbf.

GPT-5.5, reasoning off · task 06 · 3 of 3 trials

Uses the METAR wind as magnetic → **2 kt** of headwind instead of 5.5 kt. METAR winds are reported from true north.

Every transcript is in the repository

HOT TAKE

# The gap isn't arithmetic. It's *recall versus re-derive*.

The same model is right on every trial when it reasons, and confidently wrong when it answers from memory. For engineering teams, the reasoning setting matters more than the leaderboard. Ask for the governing case and every load's direction before any number, as a senior checker would. To close the gap for good, train on reversed defaults: robustness.py generates them.

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